

MD MURSALATUL ISLAM PALLOB

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EDUCATION

Dhaka International University

Bachelor of Science in Computer Science and Engineering (CGPA 3.60/4.00)

Dhaka, Bangladesh

Aug. 2021 – Oct 2025

EXPERIENCE

Brace Laboratory (bracelaboratory.com)

Founder & Chief Executive Officer

Dec 2024 – Present

Dhaka, Bangladesh

- Spearhead company operations, strategic vision, and technical architecture for software development and active research projects.
- Oversee cross-functional team workflows, ensuring high-quality product delivery while maintaining an internal policy supporting active research initiatives.

Medvisor Mobile

Lead Software Engineer intern (Part-time)

Jun 2024 – Aug 2024

Irvine, CA

- Developed and implemented machine learning models for voice and emotion stress analysis.
- Collaborated closely with Flutter, DevOps, and UX/UI engineers to integrate AI-driven features into mobile apps, aligning with project goals.

PROJECTS

Suicidal Ideation Prediction | *Python, Natural Language Processing, Machine Learning*

Aug 2025 – Oct 2025

- Built an end-to-end NLP pipeline in Python for preprocessing text data, tokenization, and feature extraction to support supervised classification of suicidal ideation from short-form text.
- Implemented and compared modeling approaches (e.g., transformer-based fine-tuning and classical ML baselines) and established a reproducible training/evaluation workflow using cross-validation and standard metrics (precision, recall, F1, AUC).
- Packaged experiments and analysis in reproducible notebooks and scripts, and documented data handling and ethical/privacy considerations to support responsible use and further research.

Infant Cry Analysis Web Application | *Python, Django, PostgreSQL, Bootstrap*

May 2025 – Oct 2025

- Developed a full-stack web application (using Django and PostgreSQL) for the detection and multi-class diagnosis of infant distress from cry recordings, providing a user-friendly interface for uploading and analyzing audio data.
- Integrated a machine learning backend capable of processing audio features (like Spectro-Derivative features) to predict and classify cries into different distress categories (e.g., pain, hunger, sleepy, healthy).
- Designed a responsive frontend using Bootstrap to display classification results, visualizations of audio features, and model performance metrics.

Infant Cry Analysis ML Models | *Python, Scikit-learn, Spectrograms, Machine Learning*

Mar 2025 – Oct 2025

- Designed and implemented a Balanced Augmented Spectro-Derivative Ensemble Framework for multi-class infant distress diagnosis, achieving high classification accuracy.
- Extracted and analyzed spectrograms and Spectro-Derivative features from infant cry audio for effective feature engineering and representation.
- Developed and evaluated various machine learning models (e.g., ensemble methods) for the classification of infant cries into specific distress types, focusing on robust performance across different datasets.

Real-time Speech Emotion Recognition | *Python, Keras, LSTM, Audio Processing*

Jun 2024 – Jul 2024

- Developed a real-time Speech Emotion Recognition system using an LSTM-based deep learning model implemented in Keras to analyze live microphone input and identify expressed emotions.
- Prepared and augmented audio datasets (RAVDESS, TESS), extracted acoustic features and spectrogram representations, and engineered time-series inputs suitable for recurrent neural networks.
- Delivered a real-time inference pipeline with continuous monitoring, achieved 87% accuracy on benchmark evaluations, and implemented silence detection to automatically stop processing for robustness.

ACHIEVEMENTS

- Winner of the Programming Competition, June 2022 — DIU.
- 2nd Place in Programming Competition at IT Carnival 2022 — DIU.
- ICPC 2021 Onsite Contestant (National Rank: 90) — BUBT.
- SUST IUPC 2023 (National Rank: 82) — SUST.
- Qualified for ICPC 2022 Onsite Contest — Green University of Bangladesh.
- NASA International Space Apps Challenge 2023 — Global Nominee; 1st Runner Up, Barishal Division, Bangladesh.
- Qualified for ICPC 2023 Onsite Contest — BUBT.
- NASA International Space Apps Challenge 2024 — 2nd Runner Up, Barishal Division, Bangladesh.
- KUET IUPC 2025 — Participated in Inter-University Programming Contest.
- UIU IUPC 2025 (National Rank: 60) — United International University.
- NASA International Space Apps Challenge 2025 — Honorable Mention; Global Nominee; Champion, Barishal Division, Bangladesh.

LEADERSHIP & ACTIVITIES

DIU Computer Programming Club

Dhaka, Bangladesh

Vice President

Sep 2024 – Mar 2025

- Oversaw club activities, including organizing coding competitions, workshops, and hackathons, to foster collaboration and skill development.
- Collaborated with the executive team to create opportunities for members to advance their programming knowledge and careers, building a strong community.
- Organized DIU CSE Fall Fest 2024.

Competition Lead

Oct 2023 – Aug 2024

- Managed diverse programming contests and authored 59 original competitive programming problems for club competitions at Dhaka International University.
- Mentored club members in competitive programming, providing structured training to prepare teams for prestigious competitions including ICPC and IUPC.

IEEE DIU Student Branch

Dhaka, Bangladesh

Secretary General

Sep 2021 – Sep 2023

- Organized seminars and fundraising events to facilitate student branch initiatives.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, JavaScript

Machine Learning & AI: Classical ML, Deep Learning (CNN, LSTM, Transformers), NLP, Audio Signal Processing, Feature Engineering, Model Evaluation

Frameworks & Libraries: TensorFlow, Keras, Scikit-learn, Django, Numpy, Pandas

Data & Tools: PostgreSQL, Git, GitHub, Docker, VS Code, PyCharm

LANGUAGES

- English - Professional working proficiency (Undergraduate medium of instruction)
- Bangla - Native

PUBLICATIONS

Accepted Papers

[1] M. M. I. Pallob, F. B. Orpita, N. Usha, N. Hasan, M. N. Tahira and M. A. Based, “Infant Cry Analysis Using Machine Learning Models and Extended Feature Selection,” *Taylor and Francis*, BIM 2025 - Accepted, Waiting for Publication.

Currently Working On

[2] A. M. Rion, M. M. I Pallob, R. Rakib, M. A. Molla and M. Hasan, “Cross-Cultural Early Detection of Suicidal Ideation in Adolescents: An Ensemble Machine Learning Approach Using GSHS Data”

[3] M. M. I. Pallob, M. A. Molla and R. Rakib, “Beyond Traditional Credit Scores: A Machine Learning Framework for Loan Default Prediction Among Unbanked Populations Using Alternative Data Signals”

[4] M. M. I. Pallob, M. A. Molla and R. Rakib, “AI-Driven Prescriptive Analytics for Hybrid Learning Mode Optimization: A Multi-Model Benchmarking and Explainable AI Framework”

— *I hereby declare that all information presented in this resumé is accurate, complete, and a truthful reflection of my qualifications, experiences, and achievements.* —

— MD MURSALATUL ISLAM PALLOB —